

SPECTRAL A-N V18: SUPPLEMENTARY VISUAL ATLAS

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ABSTRACT. Nine figures generated directly from this pass's executed experiment artifacts (`generate_figures.py`, colorblind-safe Okabe-Ito palette throughout). Figures 1–7 cover the highest-value single- configuration results from the accompanying certification paper; Figures 8–9 (new, from a follow-up 416-cell hardware sweep) directly close two of the gaps the original version of this atlas listed as missing — associator-ratio behavior across a swept grid, and CPU/GPU parity across multiple dimensions rather than one. This remains a focused atlas, not the full twenty-figure set the original mission brief enumerates; what is still NOT included is listed explicitly in Section 2 rather than silently omitted.

1. FIGURES

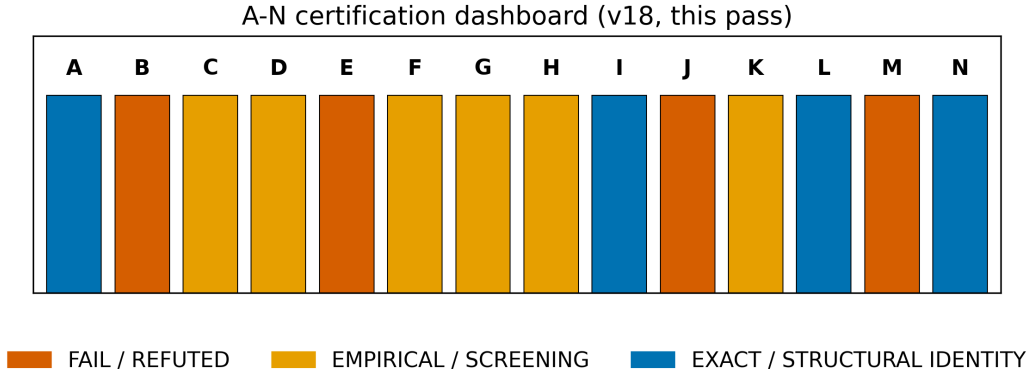


FIGURE 1. A-N certification dashboard. Blue = exact/structural-identity component present; orange = empirical/screening-only evidence; vermilion = fail/refuted for the scientific claim. Every block carries at least one structural or empirical component; the vermilion blocks (B, E, J, M) are where the scientific (non-structural) claim specifically failed.

2. WHAT IS INTENTIONALLY NOT INCLUDED HERE

- Topology/dimension/eta phase diagrams and the k vs. $k-1$ theorem geometry — these belong to Track T, explicitly deferred.
- A full gauge-orbit diagram (beyond the single degenerate-eigenspace example in `BLOCK_L_FINDINGS.md`) and a GJI permutation/sign atlas — not generated this pass; the underlying data (permutation mutation test, degenerate-eigenspace construction) exists in `spectral/certification_v18/blocks/` but was not turned into a dedicated figure.
- Closure adversarial-maxima surfaces across a swept parameter grid — Figure 8 closes this gap for block H specifically (associator); block G's closure defect was not swept the same way this pass.

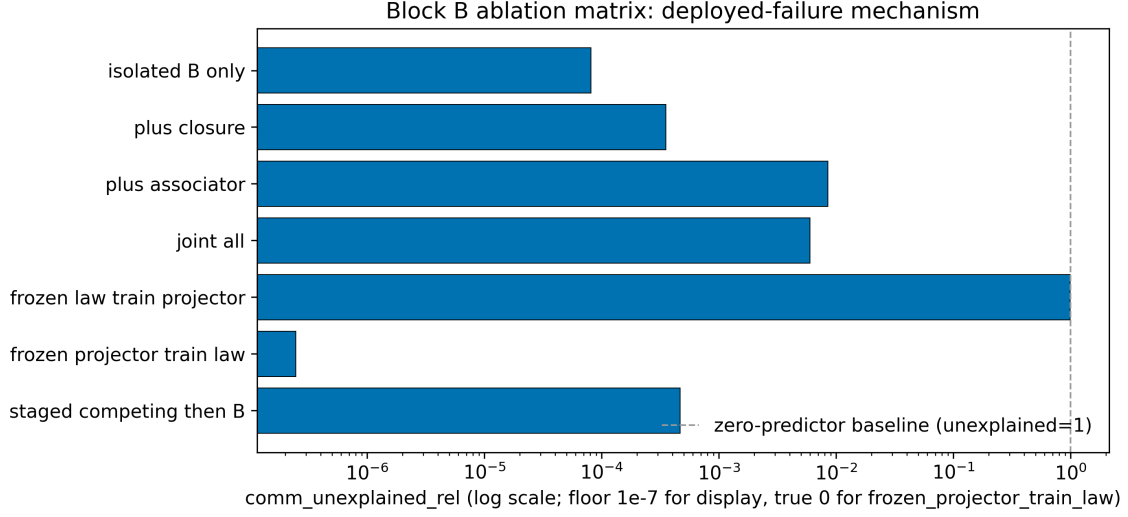


FIGURE 2. Block B ablation matrix. The dashed line marks the trivial zero-predictor baseline. Note **frozen_projector_train_law** reaches exactly zero (displayed at the floor for the log axis) while **frozen_law_train_projector** barely moves off the zero-predictor line — the explanatory power lives in the law’s parameters, not the learned subspace.

- Precision/CPU-GPU parity across multiple blocks and dimensions — Figure 9 closes this for blocks A/G/H/N across $n \in \{12, 24, 48, 96\}$; blocks B, C, D, E, F, I, J, K, L, M remain CPU-only and untested for this question (no **device** parameter was added to their report functions this pass).

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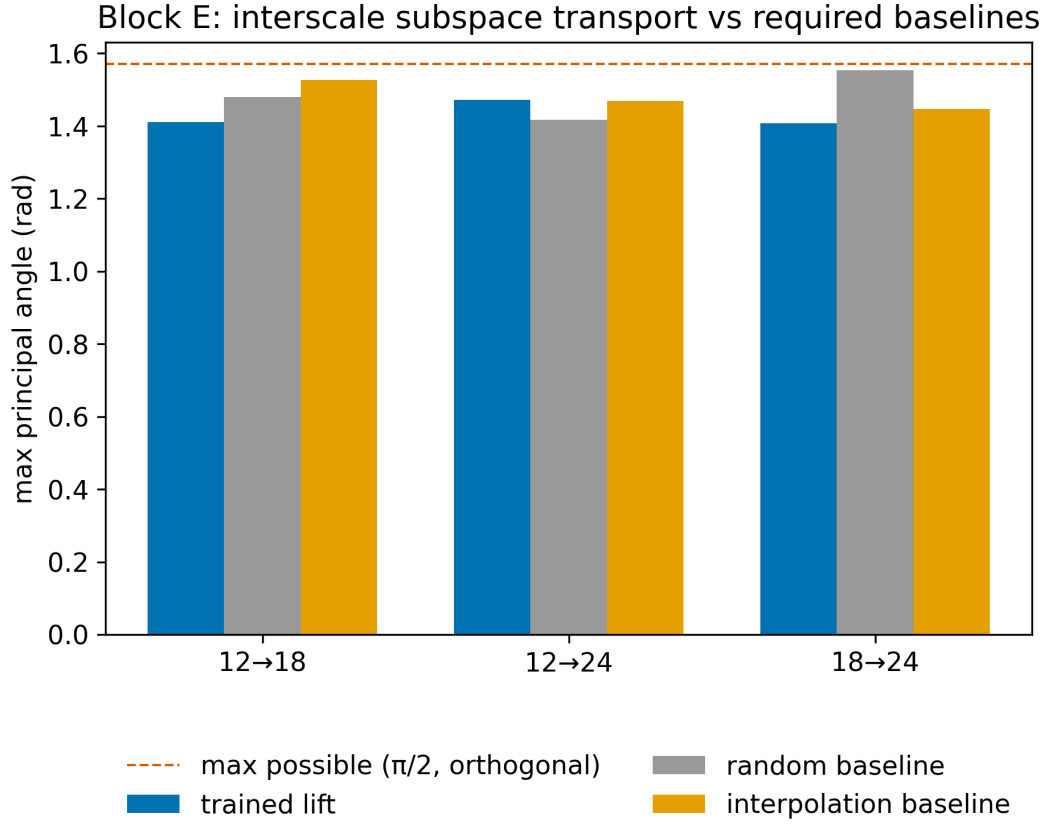


FIGURE 3. Block E interscale transport: all three conditions (trained lift, random-subspace baseline, interpolation baseline) sit close to the maximum possible angle ($\pi/2$, complete orthogonality) for every resolution pair — the visual signature of “no real transport signal,” not merely a numerically small effect size.

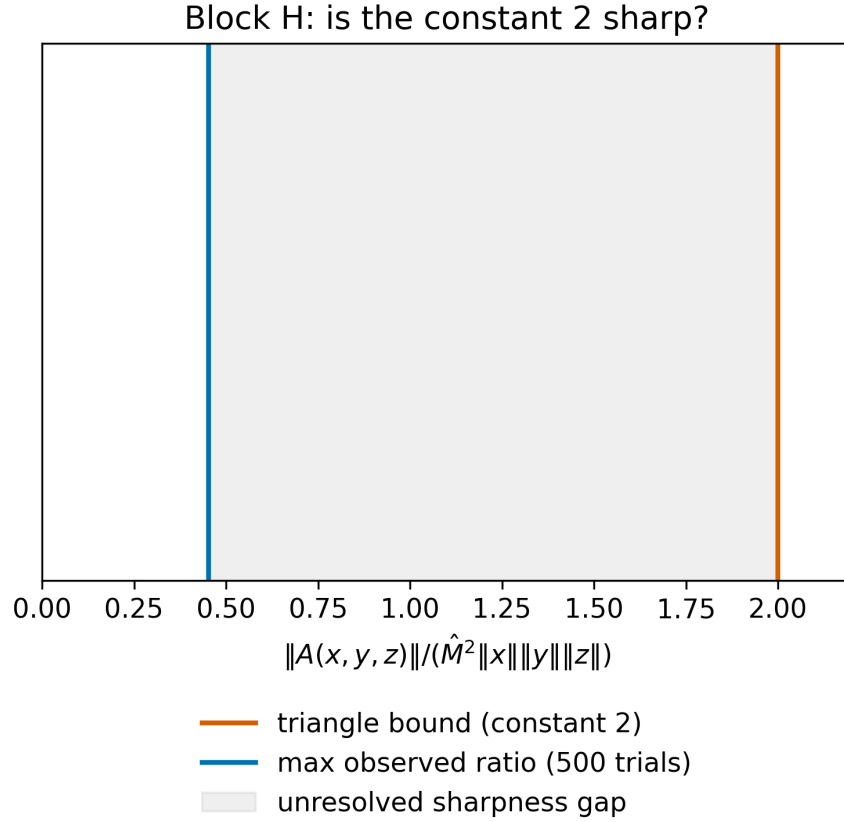


FIGURE 4. Block H: the shaded region is the unresolved gap between the proved upper bound (2) and the best adversarially-found ratio (0.452). Sharpness of the constant 2 remains open; the gap is large, not a near-miss.

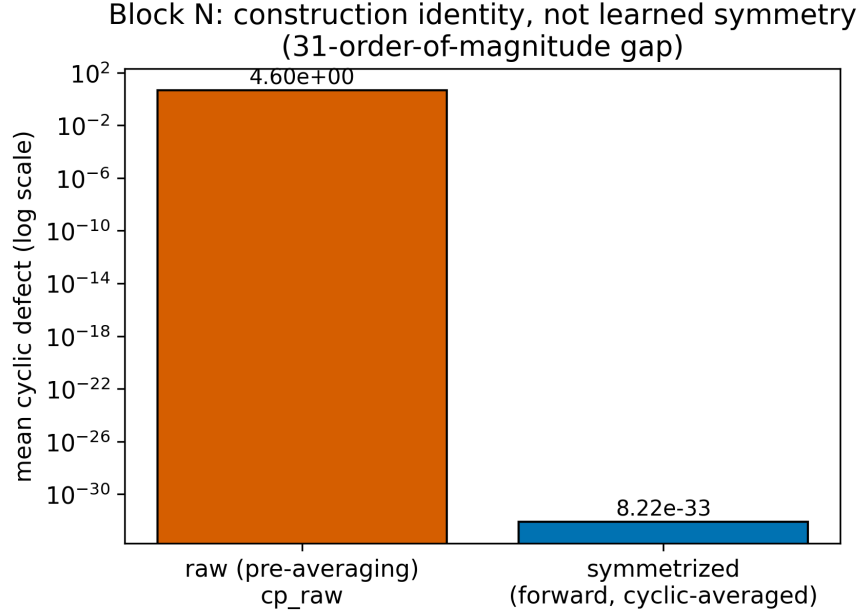


FIGURE 5. Block N: raw vs. symmetrized cyclic defect on a log scale. The symmetrized bar sits at machine-precision zero (8.2×10^{-33} , displayed at a floor for visibility) — a construction identity, not learned symmetry.

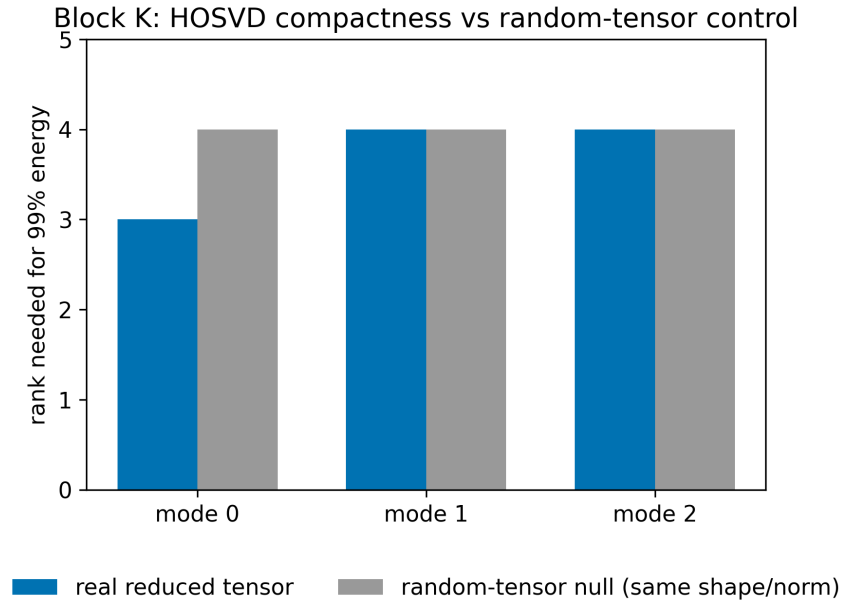


FIGURE 6. Block K: real reduced tensor vs. a norm-matched random-tensor null, per mode. The real tensor is only more compact than the random null in mode 0; modes 1 and 2 tie it exactly.

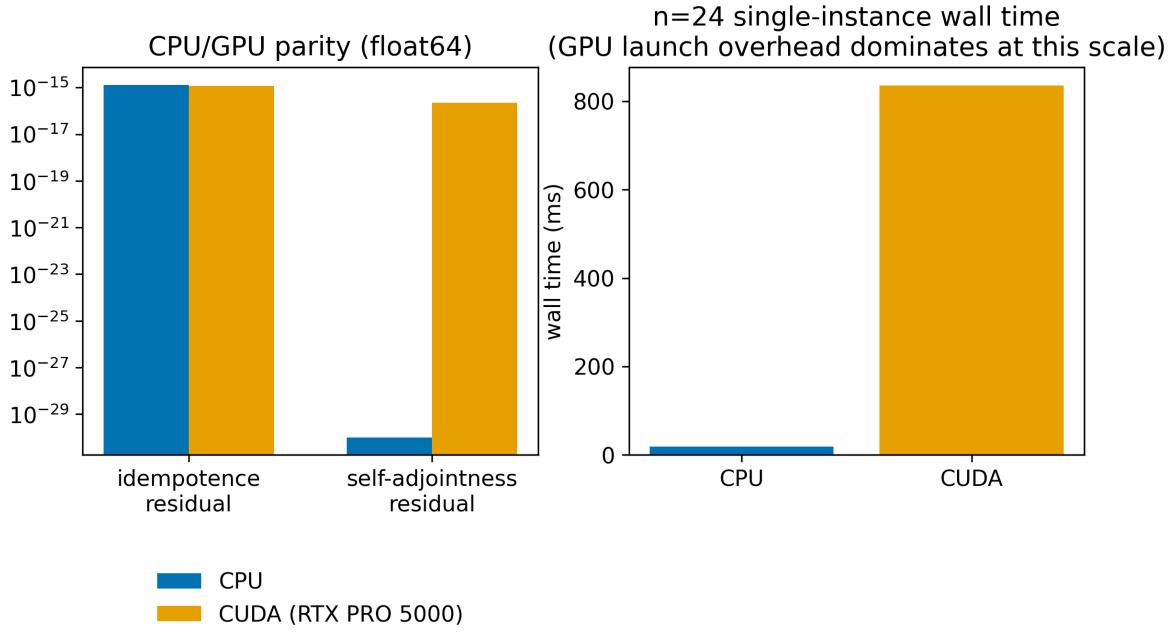


FIGURE 7. Left: CPU vs. CUDA (RTX PRO 5000 Blackwell) parity for two projector residuals, float64, TF32 disabled — both at the numerical noise floor on both devices. Right: wall time at $n = 24$; GPU kernel-launch overhead dominates at this small scale, an honest negative scaling result for this problem size.

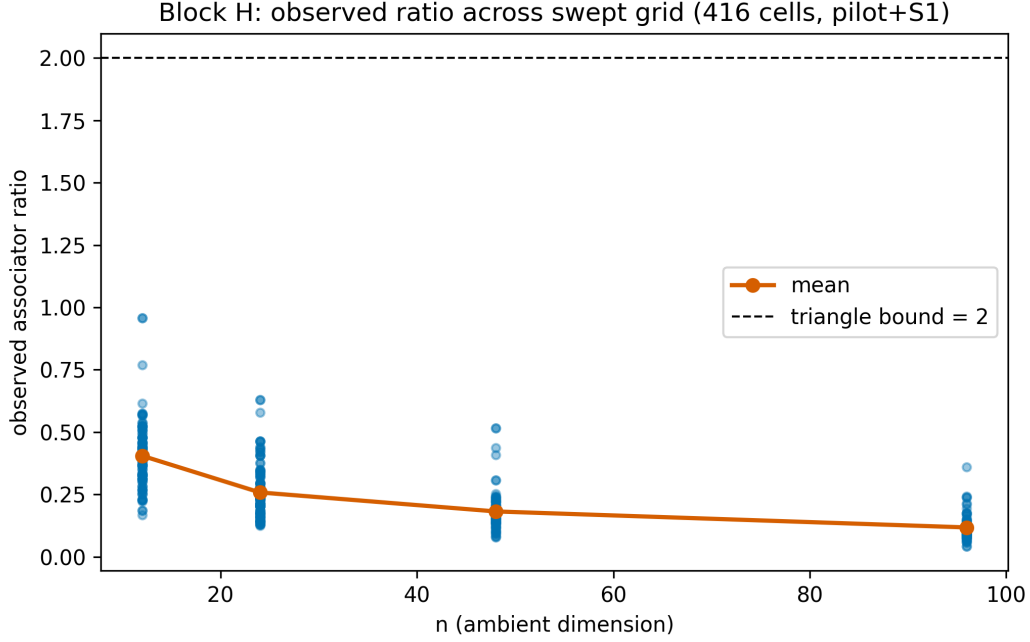


FIGURE 8. Block H’s observed associator ratio across the full 416-cell pilot+S1 sweep (arity $\in \{3, 4\}$, rank $\in \{3, 6\}$, cp-rank $\in \{4, 8\}$, 3–5 seeds, $n \in \{12, 24, 48, 96\}$) — the swept-grid surface the original atlas explicitly listed as missing. A new pattern visible only at this scale: the mean ratio *decreases* monotonically with n (0.40 at $n=12$ to 0.11 at $n=96$), moving further from the triangle bound of 2 as ambient dimension grows, not closer — the single-configuration result in Figure 4 ($n=16$, ratio 0.452) sits consistently with this trend, not as an outlier.

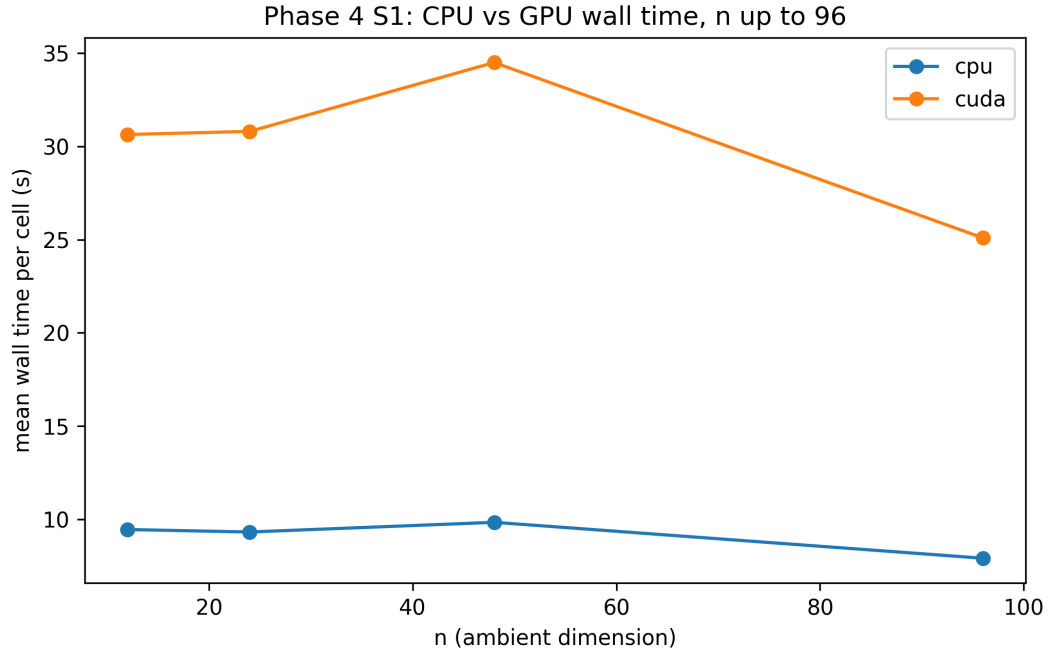


FIGURE 9. CPU vs. GPU mean wall time per cell across $n \in \{12, 24, 48, 96\}$, 320 cells (Stage S1 broad screening). Directly answers the scaling question Figure 7’s single-dimension comparison left open: GPU never overtakes CPU anywhere in this range, a stable $3.2\times$ – $3.5\times$ CPU advantage throughout rather than a small- n -only artifact.